18

Timurid-Mughal Philosopher-Kings as Sultan-Scientists

Matthew Melvin-Koushki

The successors of Amir Temür, supreme Muslim Lord of Conjunction (*ṣāḥib-qirān*), developed a distinctive form of saint-philosopher-kingship without precedent in Islamic history. Most notably, as part and parcel of their universalist-imperialist quest to transcend binaries political (caliphate vs. sultanate), cultural (Turk vs. Persian), confessional (Sunni vs. Shiʿi), religious (Islam vs. Hinduism or Christianity), philosophical (Hellenic vs. Islamic) and epistemological (manifest vs. occult) in equal measure, they fashioned themselves absolutist *astrocrats*, possessed of talismanic bodies capable of marrying heaven to earth, by means of a personal mastery of astronomy-astrology (*ʿilm al-nujūm*). In embrace of or challenge to this peculiarly Timurid model, some of the greatest rulers of the early modern Persianate world would similarly claim talismanic sovereignty.[1]

Of manifest political utility, the science of the stars had attracted the perennial interest of ruling elites since Babylonian antiquity; hewing to late antique Hellenic-Persian precedent, Abbasid and Fatimid caliphs and countercaliphs were great patrons of the same, which fostered significant Arabic advances in this Ptolemaic discipline—the invention of *conjunction astrology* most crucially.[2] But Mongol rulers seized upon astronomy-astrology with even greater alacrity, this as a scientific means of transcending the various competing cultures and scriptural religions of their vast domain. For the perfection of celes-

1 See e.g. Fleischer, Ancient Wisdom; Babayan, *Mystics, Monarchs, and Messiahs*; Moin, *The Millennial Sovereign*; Melvin-Koushki, Early Modern Islamicate Empire. On the historical evolution of the title *ṣāḥib-qirān* itself, perhaps of Middle Persian vintage, see Chann, Lord of the Auspicious Conjunction. I here use "talismanic" in the scientific sense: in Arabic and Persian classifications of the sciences, talismans (*ṭilasmāt*/*ṭilismāt*) are routinely defined as "a form of celestial magic that conjuncts celestial virtues with the virtues of certain terrestrial bodies to produce a power that gives rise to an unusual action in this world," often incorporating magic squares (sg. *wafq al-aʿdād*) to this end (Melvin-Koushki, Powers of One 148; Hallum, New Light).

2 This by Māshāʾallāh (d. ca. 200/815), and systematized and propagated by Abū Maʿshar al-Balkhī (d. 273/886); see Lemay, *Abu Ma'shar*.

 | DOI:10.1163/9789004690615_020

tial mathematics must trump the flux of mere terrestrial dogma, just as the empire of the Oceanic (Chinggis) Khan must reflect the universal sway of Tenggeri.[3] The star tables produced at the Maragha Observatory, famed as the *Zīj-i Īlkhānī*, were in turn the basis of the astronomical revolutions that would define Western scientific early modernity, from Iran to Italy, from India to Iberia.[4]

Their philhellenizing and islamicizing Timurid heirs, however, took this patronage program even further: they were the first to pursue this science within an explicitly *lettrist, Pythagorean* framework.[5] That is to say, they proposed that the Two Books, the Quran and the cosmos, be read in tandem as twin mathematical texts—whence the dual astrological-lettrist platform undergirding Timurid claims to imperial universalism, which definitively timuridized the very title *ṣāḥib-qirān*, and whence the mathematization of astronomy by the members of the Samarkand Observatory, a revolutionary development now much feted by historians of science.[6] Thus institutionalized, this same occult-scientific platform remained an effective means of performing a specifically Timurid mode of sovereignty throughout the Persianate world until at least the mid-17th century, and especially in Mughal India. It is therefore no accident that two Timurid rulers—self-styled philosopher-kings on the Alexandrian-Solomonic model[7]—were among the very few Muslim sovereigns to author *scientific* works in their own right.[8]

3 On this theme see e.g. Elverskog, The Mongols; Brack, *An Afterlife for the Khan*.

4 For intellectual-historical purposes, the West is best defined as that half of the Afro-Eurasian ecumene, incorporating the Arabic, Persian, Greek and Latin cosmopolises, where Hellenic-Abrahamic dialectics became hegemonic, and philosophy came to be pursued in mathematical-linguistic terms; see Melvin-Koushki, *Taḥqīq* vs. *Taqlīd*.

5 As Jonathan Brack observes (Mediating Sacred Kingship 275): "[The Ilkhanid] mode of sacral authority that was defined by the ruler's endowment with a unique, intuitive and even divine form of intellect that allowed him to gain access to all knowledges and sciences via direct intuition, a power expressed through the ruler's patronage and involvement in intellectual debates, retained considerable power in the Timurid period. Timurid intellectuals and court clients might be seen, therefore, as 'rebranding' this new style of intellectual kingship in a far more appealing and persuasive packaging, for example, through astrology and lettrism, than the Ilkhanid vizier [Rashīd al-Dīn]'s *kalām*-based vision of the philosopher-king."

6 See e.g. Ragep, Copernicus; Saliba, *Islamic Science*; Fazlıoğlu, Mathematical-Astronomical School; Melvin-Koushki, Powers of One. On the Two Books doctrine in its parallel Latinate iterations, see e.g. Killeen and Forshaw (eds.), *The Word and the World*; Håkansson, *Seeing the Word*.

7 It must be emphasized that both Alexander and Solomon came to stand for specifically *occult* philosopher-kingship throughout the early modern Persianate world, having been codified as such by the Ganjavi sage-poet Niẓāmī (d. 1209) in particular; see e.g. Bürgel, Occult Sciences.

8 That is, works on astronomy, member of the quadrivium, and hence "scientific" even according to the current, far narrower English usage. The somewhat anachronistic tenor of this

This contribution briefly presents the personal scientific output and patronage programs of these two Timurid philosopher-kings, both grandsons of Temür himself: Iskandar Sulṭān (r. 812–817/1409–1414) and his cousin Ulugh Beg (r. 811–853/1409–1449). The first, a hugely ambitious (if soon foiled) patron of the sciences of stars and letters, authored the preface to a state-of-the-art manual of mathematical astronomy (and likely the manual itself, which has not survived), the *Jāmiʿ-i sulṭānī* ('Sultanic Compendium'). The second, hailed by his astronomer and lettrist patronees as messianic *sultan-scientist* (*al-sulṭān al-faylasūf*), founded the Samarkand Observatory and Madrasa complex and there co-authored the crucial *Zīj-i jadīd-i sulṭānī*, aka *Zīj-i Gūrkānī* ('New Sultanic Star Tables'), which remained in use from India to England till the nineteenth century, and provided the model and benchmark for all subsequent *zīj*es produced in the Persianate world.[9] Significantly, while the intellectual-imperial projects of both men are indeed unprecedented in the Islamicate context, they may be said to islamicize—perhaps consciously—the classical model of Archytas (Ar. Arkhūṭas, d. 347 BCE), ruler of the powerful Greek city-state Tarentum and leading Pythagorean philosopher, who too was a mathematician-astronomer-king.[10] Nor do they seem to have had true successors: no subsequent Turko-Mongol Perso-Islamic sovereign is known to have personally authored scientific texts in the quest to mathematize the cosmos.[11]

qualifier aside, I use it here in reference to the natural (*ṭabīʿī*) and mathematical (*riyāḍī*) sciences generally, including the occult sciences (*al-ʿulūm al-khafiyya* or *al-gharība*) as a distinct subset of the same.

As for royal authorship of (occult-)scientific texts prior to the Timurid period, only certain Rasulid sultans of Yemen achieved distinction (albeit local) in this role, and hence mark an important 13th-century precedent to the 15th-century developments traced in this chapter. Most notable among them is al-Malik al-Ashraf ʿUmar b. Yūsuf (r. 694–696/1295–1296), the third Rasulid dynast, who authored a number of treatises on various scientific disciplines, including medicine, veterinary medicine, genealogy, meteorology, agriculture, oneiromancy, magic and especially astronomy-astrology, on which he produced the compendium *Kitāb al-Tabṣira fī ʿilm al-nujūm* ('Book of Instruction in Star Science'); he also personally constructed various astronomical instruments, including astrolabes. On the *Tabṣira* see e.g. Schmidl, Magic and Medicine, as well as the same author's contribution in this volume; on the sultan's works generally see Varisco, *Medieval Agriculture* 14–16.

9 For a summary of current scholarship on Ulugh Beg's observatory and *zīj*, see Blake, *Astronomy and Astrology*, ch. 6. On the attribution of the title *al-sulṭān al-faylasūf* to Ulugh Beg, see Fazlıoğlu, Mathematical-Astronomical School 41.

10 The definitive study here is Huffman, *Archytas of Tarentum*.

11 A partial exception is the *Nujūm al-ʿulūm* ('Stars of the Sciences'), a major though unfinished Persian encyclopedia of the sciences likely authored by the Bijapuri ruler ʿAlī ʿĀdil Shāh I (r. 965–987/1558–1579) himself, which places heavy emphasis on astrology, astral

Nevertheless, the model established by Iskandar Sulṭān and Ulugh Beg ensured that astronomy-astrology and lettrism in particular would continue to be heavily patronized by subsequent Timurid, and Indo-Timurid, dynasts (as well as by their Safavid and Ottoman competitors). This includes in the first place Emperor Akbar's grandson Shāhjahān (r. 1037–1068/1628–1657), self-proclaimed Second Lord of Conjunction (*ṣāḥib-qirān-i s̱ānī*) (i.e., after Temür, the First), who commissioned for his accession the *Zīj-i Shāhjahānī*, an updated and corrected version of Ulugh Beg's star tables of two centuries prior; tellingly, it features the first preface in the Arabo-Persian astronomical tradition to be explicitly and thoroughgoingly *lettrist* in tenor. The same feature is shared by the preface to the *Sirr-i Akbar* ('Supreme Secret'), a much-celebrated Persian translation of the Upaniṣads, the first, by Shāhjahān's son Dārā Shukūh (d. 1069/1659), concordist author of a number of other works. This famed but ill-starred saint-prince, who was executed by his more politically and militarily astute brother Awrangzēb (r. 1068–1118/1658–1707) after a failed succession bid, did prefer Sufism to astral science; but his casting of ancient Vedic *tawḥīd* in lettrist terms was similarly calculated, I argue, to signal his performance of Timurid sovereignty.

Considered together, these four works, two produced in the early 15th century and two in the early 17th, offer invaluable windows onto the profoundly intertwined evolution of early modern Islamicate intellectual and sociopolitical history, and especially history of science. A recent spate of scholarship has shed much light on that history; some relevant studies are referenced herein. But (occult) science and sovereignty as analytical categories remain very poorly integrated—due in the first place to persistent occultophobia among positivist Islamicist historians of science, who continue to disdain astrology as a hamper to astronomy, and dismiss Pythagorean lettrism as a marginal "mysticism" of no intellectual or imperial relevance, despite abundant evidence to the contrary. This abstraction of these actors from their actual sociopolitical context—defined precisely by a sweeping florescence of imperial occult philosophy, occult science and occult technology—has had the pernicious effect of rendering impossible any historiographical hypotheses as to motive. *Why* did a series a Timurid and Indo-Timurid sovereigns and scientists sud-

magic and various other occult sciences. The most profusely illustrated of all known early modern Persianate works, its unprecedentedly equalizing marriage of Ptolemaic and Vedic star science emblematizes the Oriento-Occidentalist ethos of the 'Adilshahis' Mughal rivals too, and helped inspire the epochal Sanskrit-Persian translation movement of the next century. On this pivotal work see Flatt, *The Courts of the Deccan Sultanates*, ch. 5.

denly seize upon Pythagorean cosmology and lettrist practice both, in conscious emulation of the ancients and the Imams alike? *Whence* the sheer self-confidence that allowed them to finally retire Aristotelian physics and mathematize astronomy—that is, to be consciously *modern*?[12]

As corrective, I have argued that the occult-scientific developments these and other imperial works represent must be understood within the larger narrative of the Pythagorean project, peculiar to the Islamo-Judeo-Christian West, to *mathematize the cosmos*—a project epitomized by the holy lineage *Copernicus-Kepler-Galileo-Newton*.[13] Until recently, however, Muslim thinkers and doers had been utterly elided from this *mathesis* narrative, the prevailing explanation for the origins of scientific modernity. Such extreme Eurocentrism has now been successfully challenged, and Ulugh Beg and his Samarkand astronomers (not astrologers) restored to that narrative.[14] (Given his more blatant occult-scientific proclivities, on the one hand, and his political failure, on the other, Iskandar Sulṭān remains absent therefrom.) Nevertheless, Islamicists have not yet appreciated the considerable extent to which *mathesis* was consciously carried out under specifically *Pythagorean* and hence *lettrist-kabbalist* auspices in the early modern Latinate context: these great saints of Western science were committed occultists to a man, and likewise sought to read the cosmos as a second mathematical scripture. I simply propose that the same principle be extended to their Muslim peers slightly to the east, who were often equally explicit about their Pythagorean-lettrist commitments. The double standard, of manifestly colonialist vintage, whereby Christian occultists may be cast as protomodern scientists but contemporary Muslim (and Jewish) occultists as only eternally medieval mystics must finally be forsworn.[15]

Such commitments run like a red thread through the royally authored texts presented below, tying them into a single conceptual, cosmological continuum; given the prevailing scientism or religionism of modern Islamic historiography, however, they have never yet been analyzed together. For reasons of space, I will only minimally contextualize them here, as I and other historians have done so at length elsewhere. In what follows, therefore, I will rather let our Timurid-Mughal royal authors speak for themselves, to open their projects to

12 On the respectful transcendence of ancient authorities as index of early modernity, see Melvin-Koushki, *Taḥqīq* vs. *Taqlīd*.

13 Thus Alfred North Whitehead's (d. 1947) classic statement summarizing his discussion of this lineage (*Science and the Modern World* 30): "In a sense, Plato and Pythagoras stand nearer to modern physical science than does Aristotle."

14 For representative studies see note 6 above.

15 So I argue in Melvin-Koushki, Powers of One.

comparative early modern intellectual and sociopolitical history. For much of that history remains unwritten, and most of its galaxy of Muslim authors—royal or otherwise—aggressively forgotten, due precisely to their pursuit of Occult Science as the basis for Empire.[16]

1 Iskandar Sulṭān, the First Early Modern, Pythagoreanizes Kingship

The 15th century witnessed an occultist arms race, as it were, for messianic and sacral forms of political legitimacy. It was initiated by Temür, whose messianic persona was constellated by Chinggisid, Sunni, Sufi, ʿAlid and occult-scientific inputs. But he did not himself lay claim to saint-philosopher-king status. That claim was rather made by his grandson Iskandar Sulṭān, a self-styled second Alexander the Great, whose culturally brilliant if brief reign in Shiraz and Isfahan served as model for intellectually and spiritually ambitious dynasts the Persianate world over.

Iskandar's successful performance of saint-philosopher-kingship was predicated on his identification with Imam ʿAlī and the Prophetic House, on the one hand, and his patronage of astronomers, astrologers and various other occult scientists, on the other.[17] Much has been made of Iskandar's interest in astronomy and astrology in particular; he employed scholars of the caliber of Maḥmūd b. Yaḥyā Kāshī (aka ʿImād al-Munajjimīn, 'Pillar among Astrologers'), hosted debates on a variety of subjects[18] and tasked his book workshop (*kitābkhāna*) with the production of luxury compendia on the various philosophical, mathematical, literary and religious sciences. This ambitious patronage program is epitomized by his celebrated, frequently reproduced illuminated horoscope, one of only two such to have come down to us and by far the most gorgeous, which was produced as scientific proof of his status as Alexandrian-Solomonic philosopher-king.[19] His world-conquering sovereignty was guaranteed by the kingly Sun in indomitable Taurus in the first house, an Ascendant in fiery Aries and militant Mars in fierce Scorpio in the 7th house of war; his philosophical brilliance by a massive four-planet near-conjunction—Moon,

16 On this theme see my forthcoming study *The Occult Science of Empire*.

17 See e.g. Aubin, Le mécénat timouride.

18 Binbaş, Timurid Experimentation.

19 It is now preserved in the Wellcome Library, London, as MS Persian 474. The non-illustrated horoscopes of Iskandar's half-brother Rustam (r. 817–828/1414–1425) and Ulugh Beg's son ʿAbd al-ʿAzīz (d. 853/1449) have also survived (Brentjes, Mathematical Sciences 330). On the scientific texts studied, taught or written by scholars connected to Iskandar and Ulugh Beg, see idem, *Teaching and Learning*, *passim*.

Mercury, Jupiter and Saturn—mostly in the constructive 2nd house in Gemini, that most cerebral of signs; and his literary acumen by beautiful Venus in poetical Pisces in the occult house 12.[20] With this rigorous scientific precedent, the invidious comparison of horoscopes would henceforth become a standard feature of Persianate imperial ideological warfare; even philosophers could be polemically ranked by the same measure.[21]

Such a preoccupation with astrology was hardly unprecedented or unparalleled in Western political history, to be sure, especially by the early modern period; indeed, many rulers of the much smaller states of Latin Christendom too invested heavily in astral science, and Ottoman rivals would institutionalize it at court to a far greater extent.[22] What was unprecedented and unparalleled, however, albeit little remarked in modern historiography, was Iskandar's embrace of *lettrism* (*ʿilm-i ḥurūf*) as universal—and hence inherently imperial—science. This he did at the instance of Ibn Turka of Isfahan (d. 835/1432) in particular, the foremost occult philosopher of early Timurid Iran and a leading member of the New Brethren of Purity, an informal scholarly network radiating east and north from Mamluk Cairo, who like their anonymous 10th-century namesake were ardently Pythagoreanizing occultists—but unlike their namesake were also famed scholars and influential ideologues of empire.[23] Most significantly, Ibn Turka wrote some of his most important lettrist works for the benefit of Iskandar, including in the first place his *Kitāb*

20 For a study see Caiozzo, The Horoscope.

21 Orthmann, Circular Motions; Popp, Mughal Horoscopes. The patronizing, explicitly positivist approach of the latter study—whereby astrology can never be science, only propaganda—must, however, be avoided; for our historical actors, it was clearly both. Cf. the horoscope prepared for Qāżī Burhān al-Dīn Aḥmad Oghullarï (r. 782–800/1380–1398), sultan of Sivas, perhaps the first Persian attempt to style its royal patron a saint-philosopher-king; see Karjoo-Ravary, Becoming a 'King of Islam.' For a remarkable and telling example of philosophers being ranked in similar fashion, see Ghiyās̱ al-Dīn Manṣūr Dashtakī's (d. 948/1541) horoscopic valorization of his father, Ṣadr al-Dīn Dashtakī (d. 903/1498), over his execrated rival Jalāl al-Dīn Davānī (d. 908/1502)—incidentally an Ibn Turkian lettrist-*cum*-Aqquyunlu imperial ideologue—in his *Kashf al-ḥaqāʾiq al-muḥammadiyya* ('Revealing the Muḥammadan Realities') (Bdaiwi, Late Antique Intellectualism, 169–170).

22 Reflecting the wild disproportionality in the study of Western early modernities generally, research on this theme in the western European context is far more advanced; see e.g. Yates, *Astraea*; Ryan, *A Kingdom of Stargazers*; Azzolini, *The Duke and the Stars*; Hayton, *The Crown and the Cosmos*. By contrast, Tunç Şen's unpublished 2016 dissertation, Astrology in the Service of the Empire, is the first to treat of the contemporary Ottoman context; no such studies exist on other Persianate polities, likewise frequently heavy users of astral science in conscious emulation of the Mongol-Hellenic model.

23 On this network see Fazlıoğlu, İhvanu's-Safâ; Fleischer, Ancient Wisdom; Binbaş, *Intellectual Networks*; Melvin-Koushki, The New Brethren.

al-Mafāḥiṣ ('Book of Investigations'), the first summa of Islamic Neopythagoreanism; it would continue to be celebrated and cited as such until the nineteenth century.[24] Surviving correspondence from Ibn Turka to this supremely ambitious Timurid ruler testifies to their total alliance of occult-philosophical, Pythagoreanizing purpose.[25]

But Iskandar was defeated and executed, only five years into his reign, by his much more conservative, Sunnizing uncle Shāhrukh (r. 811–850/1409–1447), who preferred Seljuq-style sultanship to occult philosopher-kingship. It is tempting to paint their contest as the triumph of a medieval Muslim ruler over an early modern one;[26] in contemporary terms, it was the triumph of *taqlīd* over *taḥqīq*, or blind imitation over empirical investigation.[27] Nevertheless, this Timurid Alexander—not unlike his Macedonian namesake, whose universalist vision likewise ended in failure, yet endured for two millennia as imperial symbol—stands as model for the new, peculiarly early modern breed of self-theosizing and hence openly occultist Muslim rulers that prospered across the post-Mongol Persian cosmopolis as a whole. Most notably, the dual lettrist-astrological platform for religiopolitical legitimation he embraced would persist as a primary driver of Timurid—and by extension Mughal and rival Aqquyunlu, Safavid and Ottoman—universalist imperialism through at least the mid-17th century.

This dual platform was first formulated, crucially, in Iskandar's one surviving work, his preface (*dībācha*) to the astronomical manual *Jāmiʿ-i sulṭānī*, which constitutes a veritable curriculum for any would-be saint-philosopher-king. Therein our royal author tells of his progressive mastery of all of the rational and religious disciplines, until he attained to the summit of the Pythagorean-Imamic 'science of Onemaking' (*ʿilm-i tawḥīd*)—which is to say, lettrism (*ʿilm-i ḥurūf*). For its imperial application to be most effective, however, this universal occult science must be paired with astronomy-astrology (*ʿilm-i nujūm*), which is likewise trained on the evolution of states in particular. For the two sciences together promise the control of one's political fate: *jafr* and astrology tell you the future; astral-talismanic magic allows you to change it.[28]

---

24 Melvin-Koushki, The Quest ch. 7. Iskandar did not live to see the completion of Ibn Turka's *magnum opus*, but clearly the latter began working on it in response to this philosopher-king's eager interest.

25 Ibid. 139–140.

26 Binbaş, *Intellectual Networks*, *passim*.

27 For an analysis and working definition of the category "early modernity" more broadly, see Melvin-Koushki, *Taḥqīq* vs. *Taqlīd*.

28 Cf. Melvin-Koushki, Persianate Geomancy 184. A comparison of Iskandar's preface with that of his Rasulid sultan-scientist predecessor al-Ashraf ʿUmar to his own manual of

While Iskandar lost the contest for control of Iran, despite such robust scientific backing, this unprecedented imperial pairing was enshrined as the backbone of Timurid imperial ideology by Sharaf al-Dīn ʿAlī Yazdī (d. 858/1454) in his *Ẓafarnāma* ('Book of Conquest'), an ornate Persian history of Temür much emulated throughout the Persiante world for centuries after, which propounds a new, astrological, cyclicist-and-monist, explicitly Pythagorean theory of history that would prove especially attractive to the equally messianic heirs of Temür in India.[29] In his analysis of Temür's horoscope, most notably, Yazdī argues that astrological proofs must always be corroborated by lettrist ones in order to ensure their scientific accuracy. In short, here Yazdī inaugurates a new form of *Neopythagorean historiography*, thereby imperially extending the project of Ibn Turka, his teacher and closest friend.[30] These New Brethren and their first Timurid patron may have failed in the short term; but in the long term they together succeeded in *rewriting the script*, as it were, for the performance of Turko-Mongol Perso-Islamic kingship—(occult) scientifically.

In view of its exceptional importance for mapping subsequent Persianate scientific and imperial developments, and for comparative study of parallel Latinate ones, I translate Iskandar's preface in full:

> *In the name of God, all-merciful, ever merciful—His help we seek.*
>
> Abundant encomium and limitless praise be to the Sultanic Presence, the degree of ascension of whose Sun of power and supremacy cannot be accurately measured by astrolabe or gauged by the senses—*And they measured not God with His true measure* (Q 6:91, 22:74, 39:67):[31] a Benefactor whose infinite starlike graces and planetary blessings are unchartable in the sky of enumeration by the scopes of thought and conjecture—*Though you number them, you will never count God's blessings* (Q 16:18)—, nor has the advent of the Hermes that is intellect and

astronomy-astrology, the *K. al-Tabṣira* (see note 8 above), provides a highly revealing index of the dramatic epistemic and political shifts in the Islamicate heartlands between the 13th to 15th centuries. Most notably, while both are equally concerned with the occult-scientific applications of *ʿilm al-nujūm*, the former could plausibly develop a physics-metaphysics of Mongol-style imperial universalism; the latter, naturally, could not.

29 And especially threatening to contemporary scholarly opponents like Ibn Khaldūn, whose own cyclicist theory was clearly meant as a dualist-materialist retort to Yazdī's; see Melvin-Koushki, In Defense 393–394; idem, Imperial Talismanic Love.

30 On the nature and considerable, albeit partial, influence of Yazdī's model, see Melvin-Koushki, Toward a Neopythagorean Historiography.

31 My Quranic translations throughout are adapted from that of Arberry, *The Koran Interpreted*.

understanding and the Ptolemy that is imagination (*khayāl*) and intuition (*vahm*) served to fix the tally of praise due Him—*Few of My servants are thankful* (Q 34:13)—, nor have the demonstrations of the geometricians of thought and the computers of signs explicated the exalted principles of His fabrications—*Your gaze comes back to you dazzled and strained* (Q 67:4); a Sage who opened the pathways whereby [elemental] progeny (*mavālīd*) may be derived from the marriage of higher active Fathers with lower receptive Mothers—*He directs the affair from heaven to earth, then it ascends back to Him* (Q 32:5). He inscribed the form of the robe of honor that is being human (*khilʿat-i insānī*), or mediator of the mainstream of creation and center anchoring the circumference of knowledge and insight, with signs encrypting the secrets of rulership and dominion—*We indeed created man in the fairest form* (Q 95:4)—, and made the horoscope of [humanity's] astonishing nature site of the *star-fallings* (Q 56:75) of spiritual and physical perfections and arrowtargets of earthly and heavenly felicities; and having adorned the Transcript that is [the human individual's] perfection of being (*nuskha-yi vujūd-i kāmil-ash*)—a comprehensive compendium of sultanic signs and definitive collection of divine and kingly qualities—with the proem of *And He taught Adam the names in their entirety* (Q 2:31), He raised his noble standard to the height of *I am setting in the earth a vicegerent* (Q 2:30). The dawnings of the depths of his thought thus became horizon to the day-rays of truth, and by his purpose were raised forth from the cosmic showcase of becoming the glad tidings of eternity's morn, for as long as time shall persist:

Every day does the Sun rise from that collar—
from within his shirt, my God, what an auspicious Ascendant![32]

He is lord of the three [elemental] progeny (*musallasa-yi mavālīd*)[33] and sublord of *All is He* (*hama ū-st*)[34]—indeed, were it not accounted unbelief to divulge the secret of lordship (*rubūbiyyat*), I would have declared of him too that *All is he*! For *Glory be to Him Who created Adam in the image of the All-merciful.*[35] *He is the First and the Last, the Manifest*

32 Kamāl Khujandī, ghazal: *dīda dar ʿumrī zi rūy-at bā khayālī fāʾiʿ ast.*

33 I.e., minerals, plants and animals, all products of the four elements.

34 This Persian declaration is often cited as popular motto of the 'oneness of being' (*waḥdat al-wujūd*) school of Ibn 'Arabī's followers in the east.

35 Cf. Bukhārī, *Ṣaḥīḥ, kitāb al-istiʾdhān*, bāb 1; Muslim, *Ṣaḥīḥ, kitāb al-birr wa-l-ṣila wa-l-ādāb*, bāb 32; etc.

and the Occult; He has knowledge of everything (Q 57:3).[36] *Like Him there is nothing; He is the All-hearing, the All-seeing* (Q 42:11).

And boons of pure blessing from the Giver of all gifts be scattered upon the glorious and rapturous garden illumined by that Lamp, yea, the salvific herb of the garden that is the Holy Family. For he is the foremost part (*juzv*) of the comprehensive scripture that is existence,[37] blazoned Moon of the mansions of munificence and generosity, the Greater Light (Sun) in the Midheaven of majesty, the Greater Benefic (Jupiter) in the house of its exaltation that is prophethood and messengerhood—

The prophets, for all their gravity,
are altogether a mere zero next to this integer.
So what if they came before him?
That Zero precedes One only makes the latter greater[38]—

who reflects abroad the lightbeams of revelation and inspiration, who draws the solicitous sights of grace and blessing, who has proved every facet of divine law and explicated the rigor of its judgments:

Muḥammad—from eternity to eternity all that is
is limned with the ornament of his name.

The most perfect blessings be upon him, and the most encompassing greetings be unto him.

Further: In every second—nay, millisecond (*ʿāshira*)—of the [degrees of the] orbits that shall repeat their cycles within the sphere of the zodiac and all the other heavenly spheres until the Day of Doom (*yawm al-qarār*), a thousand thousand praises and blessings from the Protector (*muhaymin*), the Peace (*salām*), are laid before the pure tombs and the fragrant gravesites of that preeminent one and his exalted Family (*ʿitrat-i nāmdār*), stars of the firmament of guidance and the sign of election—and most especially the chosen one honored with the declaration *your flesh is my flesh, your blood my blood*[39] and appointed by the holy verse *God only is your friend* (Q 5:55), proem (*dībācha*) of the compendium of

36 Enshrining the central lettrist principle of the *coincidentia oppositorum*, this verse functioned as a veritable motto of the New Brethren (Melvin-Koushki, In Defense 387; and see below). Its deployment is therefore highly significant here, and bears out Iskandar's claim to have mastered lettrism at the instance of Ibn Turka.

37 This is in reference to the traditional division of the Quran into 30 equal parts for purposes of recitation.

38 Sanāʾī, *Ḥadīqat al-ḥaqīqa va sharīʿat al-ṭarīqa* bk. 3: *andar gushādan-i dil-i vay*.

39 This hadith, needless to say, is strongly Shiʿi in tenor, appearing multiple times in Muḥammad-Bāqir Majlisī's (d. 1111/1699) comprehensive *Biḥār al-anwār* ('Seas of Lights') as such (23:126, 38:248, 40:203, 99:106).

the congregations of the righteous and master index of the star tables (*fihrist-i jadāvil-i zījāt*) of sciences and mysteries,

Commander of the Faithful,
mouthpiece of truth,
triumphant over every hero,
Imam of all Muslims by right:
ʿAlī son of Abū Ṭālib.

O God, bless and keep Muḥammad and Muḥammad's House abundantly and perpetually for as long as day differs from night and morning follows evening and the pole stars spin.

To proceed: Thus says the servant of God Most High and lord and ruler of His servants, Iskandar son of ʿUmar Shaykh (God pardon and be pleased with them both): Inasmuch as boundless divine solicitude chose this wretch from among all His creatures and arrayed the stature of his aptitude and noble purpose with the robe of physical and spiritual caliphate, He adorned the decree of his ascendancy with the seal of *the sultan is God's vicegerent on earth* (*al-sulṭān khalīfat Allāh fī l-arḍ*)[40] and illuminated the niche of his nature with the lights of the secrets of *My saints are under My domes—none knows them but I*,[41] making his body (*ẓāhir-ash*) the site of manifestation for the intricate virtues of kingship (*salṭanat u pādshāhī*) and his mind (*bāṭin-ash*) revelatory of intricate sciences and metaphysical knowledge: praise be to Him, yea, praise be to Him!

With heart and soul do I glorify a Lord who has made
a lump of dirt worthy of His graces!

According to the dictum *If you are thankful I will surely increase you* (Q 14:7), this wretch has discharged the obligations of thanksgiving for such limitless beneficence by improving cities and easing people's lives so assiduously and to such an extent that the effects of his labors shall remain inscribed on the pages of the world and the transcript of day and night until the end of time.

Concurrently with his rigorous enactment of the prescriptions of justice and equity and scrupulous attention to the care of his subjects

40 Cf. Q 2:30, 38:26. On the debate over and contest for the royal title *khalīfat Allāh* in the 15th and 16th centuries, see Markiewicz, Crisis of Rule.

41 This *ḥadīth qudsī*, significantly, is cited by Sayyid Sharīf Jurjānī (d. 816/1413) in his *al-Taʿrīfāt* ('Definitions') under his definition of the Malāmiyya, or Blameseekers (285–286). On Jurjānī's connection to Iskandar's court, see Binbaş, Timurid Experimentation; Melvin-Koushki, Imperial Talismanic Love.

(*raʿiyyat-parvarī*)—for *the justice of one hour is better than seventy years' worship*—, he devoted his best hours to the acquisition of varieties of apodictic knowledge (*ʿulūm u maʿārif-i yaqīnī*) and the hoarding of spiritual virtues and attainments (*fażāyil u kamālāt-i ḥaqīqī*), the capital of eternal felicity and vessel of everlasting blessedness, continually directing the reins of his endeavor toward this noble goal; and as a consequence of the guidance vouchsafed him by divine solicitude (the radiance of whose effects shine forth clearly in all of his states and circumstances *like the Sun at heaven's zenith*[42]) he quickly encompassed all of the traditional and rational sciences in both fundamental principles and applications, and in his analysis of the various issues of each attained such a level of proficiency that even seasoned authorities on the subject could not begin to match. Indeed, according to the dictum *the lords of nations are inspired men* (*arbāb al-duwal mulhamūn*), in each of these fields he proposed original questions and made rare observations, all wholly unprecedented, and in each technical craft (*ṣanʿatī*) invented precise wonders and wonders of precision: *That is God's bounty; He gives it to whom He will* (Q 5:54, 57:21, 62:4).

Now those who possess insight hold it to be established that the purpose behind the creation of man (and specifically the creation of him to whom He said *Were it not for you I would not have created the spheres*,[43] [namely, Muḥammad]) was so he could know God Most High and His Names and exalted attributes, acts and effects. The Sufis (*ahl-i taṣavvuf*) term this [knowledge] the science of divine oneness (*ʿilm-i tawḥīd*), and seek to attain it through unveiling and direct experience (*kashf u ẕawq*). [For their part], the inquiring among scholars of the formalized sciences (*muḥaqqiqān-i ʿulamā-yi rusūm*) who are not content with simple unreasoning imitation raise their heads to rap on the door of this science with the arm of intellect (*ʿaql*) and the hand of reflective thought (*fikr*): *Those—they are called from a far place* (Q 41:44). Among such individuals, one group champions dialectical theology (*kalām*) as the proper means to attaining this felicity, while the other pins their fortunes on metaphysics (*ḥikmat-i ilāhī*).

Everyone makes it to this alley by some means or other.

Thus with perfect desire and sincere purpose I first took up the science

42 This hemistich (here slightly truncated) appears in the diwans of both Ibn al-Rūmī and al-Mutanabbī.

43 This hadith is not included in the canonical Sunni collections, but much cited by Sunni and Shiʿi authors nonetheless.

of divine oneness, studying the results obtained through the unveilings of saints and great men among the righteous forebears (God be pleased with them all), each of whom expressed his useful findings according to his own outlook and experience, until I understood them *in toto*; I also absorbed the fragrant discourse of the great thinkers of our own day (God multiply them and lengthen their lives), whether in person through discussion or through the perusal of their treatises and books. [To this end] I gained an especially deep and thorough knowledge of lettrism (*ʿilm-i ḥurūf*), the appearance of which is a peculiar feature of the glorious age of our holy Prophet, and by virtue of the raptures born of divine graces—and *a rapture from God is equal to the work of the Two Weights*[44]—I became acquainted with many rare secrets of divine unity and numerous intricacies of metaphysical knowledge were revealed to me that are not recorded in the works of previous thinkers or mentioned by any contemporary: *And as for your Lord's blessing, declare it* (Q 93:11). After learning all the technical vocabulary of the scholars, then, I became wholly familiar with the positions of every category of learned individuals—litterateur, traditionists, jurists, theologians, philosophers, mathematicians (*ahl-i riyāżī*), experts in the mathematical sciences (*arbāb-i taʿālīm*),[45] and so on—in all their variety.

Now after the science of divine oneness, which is necessarily the most fundamental objective, there is no discipline more important or beneficial than astronomy-astrology (*ʿilm-i nujūm*). All classes of people, from low to high, and most particularly governors, leaders, sultans and kings, have need to know [somewhat of] this science, the more fully the better; the truth of this statement is self-evident and requires no demonstration. Having therefore set aside various times to study this noble art, I became preoccupied with mathematical astronomy (*ʿilm-i hayʾat*), the foundation of this larger science.

I then turned my attention to [its practical applications, including] operations based on astronomical tables (*aʿmāl-i zīj*) and the method of deriving calendars (*taqāvīm*) and the Ascendant in nativities (*ṭāliʿ-i mavālīd*) and the revolutions [of the years of nativities] (*taḥāvīl*)[46] as well

44 I.e., humanity and jinn.

45 *Taʿālīm* (= Gr. *mathēmata*) here denotes the quadrivium, i.e., arithmetic (*ʿilm al-ʿadad*), astronomy (*ʿilm al-hayʾa*), geometry (*ʿilm al-handasa*) and music (*ʿilm al-mūsīqā*).

46 I.e., *taḥāvīl-i sinī l-mavālīd*. Cf. Abū Maʿshar's *Kitāb Taḥāwīl Sinī l-ʿĀlam* ('Book of Revolutions of the World-Years').

as all the further operations and questions associated with these, deriving astrological judgments (*istinbāṭ-i aḥkām*) [on this basis] as their natural product.

[In the process of so doing] I also collected most of the writings of the ancients and latter-day authorities on this science. However, I did not find among them a complete account of its divisions that would be sufficient for all a student's needs. For this reason I have compiled and organized the present comprehensive account that encompasses all the concerns and objectives of these techniques, and accordingly entitled it the *Jāmiʿ-i sulṭānī* ('Sultanic Compendium'). It is [purely functional and] not particularly polished, being generally improvised.

Since of all the astronomical tables on which the above operations depend the *Zīj-i Īlkhānī* ('Ilkhanid Star Tables') are the most correct and most reliable, I have made them the basis of the practical section of this work. And given that their author (may he rest in peace) used a high register and difficult style in composing them, in the case of each operation I provide an appropriate example to clarify the point.

[Finally], because the foremost of the science's subsets is the science of the configuration [of the stars] (*ʿilm-i hayʾat*), it has been dealt with at the beginning of the book in summary form over twenty chapters.

I pray the infinite mercy and boundless grace of God Most High that this praiseworthy undertaking shall be an agent of blessing in this world and a reason for clemency in the next. My success is only through God; on Him do I rely, to Him do I turn.[47]

2 Ulugh Beg, Sultan-Scientist, Mathematizes Astronomy

Shāhrukh was a conservative ruler with little interest in science; but his son Ulugh Beg, like his rival Iskandar Sulṭān, was decidedly and consciously a modern one, and scientifically achieved what his more politically ambitious cousin had merely hoped to. As such, he has been celebrated by historians of science as a figure virtually unique in Islamic history: equally a sultan and a scientist, and therefore with personal control of the necessary imperial resources to conduct the most successful astronomical observation program since that enshrined in Ptolemy's *Almagest*—and the first in Western intellectual history to push for

47 *Dībācha-yi Jāmiʿ al-sulṭānī* 207–211. The fact that this preface is only preserved in Yazdī's collected short writings is here highly significant.

the total mathematization of astronomy, thereby prefacing the Copernican and then Newtonian revolution of the next three centuries.[48]

What the same historians have not inquired into, however, is the cosmology undergirding and driving Ulugh Beg's revolutionary *mathesis* project. Based on circumstantial evidence, I have elsewhere argued that that cosmology was no less Neopythagorean than that of Iskandar and the New Brethren.[49] For the *Zīj-i jadīd-i sulṭānī*, like all *zīj*es, is designed in the first place for astrological use;[50] Ibn Turka cultivated Ulugh Beg's patronage by dedicating an important lettrist work to him and declaring the Samarkand Observatory the realization of his own hopes, and Yazdī, mathematician-lettrist as well as historian-litterateur, even worked there for a time;[51] Qāżīzāda Rūmī (d. ca. 835/1432), second director of the observatory, was a friend to Ibn Turka from childhood, and the two carried on a warm correspondence;[52] ʿAbd al-Qādir Rūyānī Lāhījī (d. 925/1519), student of Ulugh Beg's preeminent astronomer ʿAlī Qūshchī (d. 878/1474), authored works on astronomy and geomancy (*ʿilm al-raml*) both—the latter conceptualized during this period as a quintessentially Pythagorean divinatory science, and defended by Yazdī as such in specifically lettrist terms;[53] and the sultan-scientist himself was reported to have regularly consulted geomancers, once, memorably, in the company of Qūshchī.[54] (The fact that a Safavid-era copy of the *Zīj-i jadīd-i sulṭānī* includes a geomantic procedure on the flyleaf is here suggestive.[55]) Similarly, it is no coincidence that Ibn Turka completed his *Book of Investigations*—which issues a clarion call precisely for the mathematization of the sciences—in 1420, the same year construction of the Samarkand Observatory was begun.[56]

Ulugh Beg and his Samarkand team alacritously answered this call, alive to the Pythagorean cosmological framework whence it was issued. At the same time, this son of Shāhrukh was constrained by the prevailing political atmosphere of Shāhrukhid Iran—featuring the persecution of lettrists like Ibn Turka on the pretext of association with the anarchic Ḥurūfiyya[57]—to be consider-

48 Blake, *Astronomy and Astrology* 93.
49 Melvin-Koushki, Powers of One.
50 Şen, Astrology in the Service, *passim*.
51 Melvin-Koushki, The New Brethren; Binbaş, *Intellectual Networks* 65.
52 Melvin-Koushki, The Quest 50–51, 146–147.
53 Idem, Persianate Geomancy 167–168.
54 Şen, Astrology in the Service 177 note 29.
55 Tehran, MS Majlis 72s, 1b–2b (the geomantic operation, for determining the waiting period to remarry, is on f. 1[a]).
56 Melvin-Koushki, The Quest 97–99.
57 Binbaş, The Anatomy; Melvin-Koushki, The Quest ch. 8.

ably more modest and conservative in the presentation of his unprecedented project.[58] In the preface Ulugh Beg penned for his watershed *zīj*, he claims neither saint-philosopher-kingship nor *ṣāḥib-qirān*-hood, and does not explicitly invoke the science of letters; his account of his self-training as philosopher-king seems perfunctory in comparison to Iskandar's, and moves quickly to acknowledge his fellow mathematian-astronomers Ghiyāṡ al-Dīn Jamshīd Kāshī (d. 832/1429), Qāżīzāda Rūmī and ʿAlī Qūshchī as equal partners in the three-decades-long project.

This feature has allowed positivist historians of science to simply ignore the ostensibly rhetorical flourishes of the preface in favor of data-mining. And yet those flourishes contain a code that becomes decipherable when restored to its proper cosmological, generic and social contexts.[59] For Ulugh Beg explicitly couches his project in terms of the mathematical decoding of the Two Books; he even confesses his investment in the 'arcane among the sciences' (*ghavāmiż al-ʿulūm*)—a standard Arabo-Persian synonym for the occult sciences (*al-ʿulūm al-khafiyya* or *al-gharība*) as a distinct subset of the natural and mathematical sciences—without explicating further. Most radically, even paradoxically, he seems to subjugate the rational sciences to the religious—yet immediately asserts, in classic Mongol fashion, astronomy-astrology to be the true universal-imperial science transcending all temporally bound and hence changeable religions, including Islam. This is hardly a thesis that would have been acceptable to the Sunnizing and orthodoxizing Shāhrukh.

Read in its originary, highly competitive intellectual-imperial context, therefore, even the relatively formulaic opening of Ulugh Beg's preface provides clues as to the cosmological motivation of this, the greatest of Muslim sultan-scientists. (His treatise on determining the sine of one degree has unfortunately been lost.[60]) I translate it too in full:[61]

58 Cf. Edward S. Kennedy's conclusion that Ulugh Beg was more able administrator than scientist (The Heritage).

59 The coding of prefaces is epitomized in the rhetorical device termed *barāʿat al-ihtilāl*, or the encoding of one's major theme in the opening lines of a work or letter. For an example and analysis, see Melvin-Koushki, The Delicate Art.

60 An extant treatise so titled, *Risāla fī Istikhrāj jayb daraja wāḥida* ('On Determining the Sine of One Degree'), has been attributed to Ulugh Beg, though it appears to be rather the work of Qāżīzāda on the basis of a similar treatise by Kāshī. But Ulugh Beg states in the *Zīj* that he too produced a separate treatise on the subject, which, like Kāshī's, is no longer extant; see Savādī, Naqdī bar istidlāl-i Ruzinfild; van Dalen, Ulugh Beg.

61 Cf. the (often inaccurate) translation of Louis-Pierre-Eugène Amélie Sédillot (d. 1875), one of the first orientalists to edit the *zīj* (*Prolégomènes* [1853] 1–6)—its causual inaccuracy reflecting his earlier dismissal of the preface in his edition of the text as "un échantillon de style oriental, presque intraduisible" (*Prolégomènes* [1847] cxxxvj).

In the name of God, all-merciful, ever merciful.

Blessed be He who has set in heaven the zodiacal constellations, and has set among them a lamp, and an illuminating Moon! It is He who made the night and the day a succession for whom He desires to remember and whom He desires to be thankful (Q 25:61–62)—the Lord of Kingship whose dawnlight that is *We have set a blazing lamp* (Q 78:13) is the kindling of His wisdom and whose blazon that is *We have cast down from the clouds copious water* (Q 78:14) is the standard of His will—that Mighty One whose wisdom and will having been observed to (*rāṣid*) and erected by the seven levels of heaven, without mediation of number or physical aid, by means of the might of *He raised up heaven without pillars* (Q 13:2), He illuminated the manuscript pages that are the spheres with pearlescent forms—many thousand scintillating stars both fixed and planetary—with the brush of *We have embellished the lower heaven with the embellishment of the stars* (Q 37:6), and in accordance with the dictum *And the earth—after that He spread it out* (Q 79:30) beautified, by the mere command *Be! and it is* (Q 2:117, et al.), the face of the desert and the sown with every manner of brilliant invention and artifice.

From out the matter of the [four] elements and their [three] progeny[62] He then chose man; having inscribed the distinguishing writ *We have created man in the fairest form* (Q 95:4) on the page of his being, He exalted him to the sublimest degree of *We have ennobled the children of Adam* (Q 17:70). And in accordance with the dictum *And those messengers, some we have preferred above others* (Q 2:253), He further cast the robe of His preference about the holy shoulders of one Messenger in particular, the sleeves of whose majesty are embroidered with *Were it not for you I would not have created the spheres*.[63] Pursuant to *My companions are like the stars: you shall be guided by whomsoever of them you follow*,[64] He then designated his sons and grandsons and companions and dear ones[65]—stars of the heaven of leadership, suns of the earth of munificence—with the royal robe of the caliphate and the ornament of the imamate. *O believers, do you bless him, and pray him peace* (Q 33:56)!

62 See note 138 above.

63 The same hadith is cited by Iskandar; see note 42 above.

64 Likewise a hadith not in the canonical Sunni collections.

65 Thus Sédillot's edition (287), as well as MS Majlis 72s, which give the series *awlād u aḥfād u aṣḥāb u aḥbāb*; a less explicitly Sunni formulation is given in MS Majlis 16397, which drops the *aṣḥāb*.

To proceed: Thus says the most wretched of God's servants and the one most needful of God the Besought, Ulugh Beg son of Shāhrukh son of Temür Güregen (God gladden his being and fulfill his hopes): Despite manifold distractions and constant labor in guaranteeing the welfare of the nations and the ways and means of mankind, I nevertheless—following the dictum *A man soars on the wings of his high purpose*—devoted the entirety of my purpose to and trained my utmost aim on the attainment of the citadels of learning and the encompassing of all its noble and ennobling leavings. I therefore turned my reins of virtuous travail and extreme effort toward the acquisition of scientific truths and philosophical intricacies, till that divine aid attended this wretch, and—according to the dictum *Whoso seeks determinedly shall find*—with the pen of intelligence and reed of intellection I explicated sciences occult (*ghavāmiż-i ʿulūm*) and disciplines technical (*daqāyiq-i funūn*)—above all mathematical astronomy (*ʿilm-i hayʾat*) and the [affiliated] philosophical sciences (*ʿulūm-i ḥikmī*), whose sanctuary is not soiled nor roiled with the obscuring dust that is the changing of religions and the contradictory dogmas of time. Given that the Creator (glory be to His name!) had from out the treasury of His universal munificence that is *There is naught but its treasuries are with Us, and We send it not down but in a known measure* (Q 15:21) so greatly honored this poor servant with this supreme boon, he purposed to have the moral of

Our works tell of us—
look to our leavings after we've gone

written upon the record of the wonders of the age.

Having thus raised aloft the standard of glory atop the dome of the circling sphere, he therefore appointed astronomers for this work. It began with the collaboration of the masterful scholar of scholars, our lord Ṣalāḥ al-Milla wa-l-Dīn Mūsā, known as Qāżīzāda Rūmī (God's mercy and forgiveness be upon him), erector of the standards of erudition and philosophy, traveler on the paths of free inquiry (*taḥqīq*) and rigorous precision; and our most eminent lord Ghiyāṡ al-Milla wa-l-Dīn Jamshīd (God cool his resting place), glory of the sages of the world—the illuminating mind of each being a guiding candle unto the community of scholars, or rather an oracular world-revealing cup spreading learning abroad. In the initial stages of the work, however, our lord Ghiyāṡ al-Dīn Jamshīd (may he rest in peace), righteous and forgiven, having heeded the call *Answer God's summoner* (Q 46:31), journeyed forth from the abode of deception that is the world to the abode of bliss that is paradise; before this project was well underway that master (God reward his efforts!)

> was joined to the mercy of the Creator. Thereafter this great and arduous undertaking was finally accomplished with the collaboration of [this wretch's] esteemed son[66] ʿAlī b. Muḥammad Qūshchī—who had already in his youth attained prominence in various sciences to such a degree that one could be sure that the renown of his deeds would very shortly propagate to all corners of the world, God Almighty willing—by means of divine solicitude and infinite effluxion of grace. The stellar movements that became known [to us] through observation and testing are set down in this book.[67]

3 Shāhjahān, Lord of the World—Aesthetically and Scientifically

The imperial Pythagorean contest thus initiated by these two Timurid philosopher-kings in the 15th century reached its full intensity in the 16th and early 17th with the approach, then arrival, of the Islamic millennium. Yet the aspiring millennial sovereigns who were Temür's Mughal heirs or Ottoman, Aqquyunlu, Safavid and Uzbek rivals did not themselves author scientific works, occult or otherwise; but they patronized their production to at least the same extent. As Islamicist historians of science have now shown, the narrative of Islamic scientific decline after the 11th century must be rejected as the colonialist-orientalist chestnut it is. The long neglect of early modern Islamicate intellectual and sociopolitical history, however, means that even the most seminal scientific works have yet to be edited and published, much less studied.

One such work is Farīd al-Dīn Masʿūd b. Ibrāhīm Dihlavī's (d. 1039/1630) *Zīj-i Shāhjahānī*, based on that of Ulugh Beg, which, as noted, is the first Arabo-Persian instance of the genre to be framed in explicitly lettrist-Pythagorean terms—thus making explicit the relatively implicit Pythagorean tenor of its model.[68] Most notably for our purposes here, Farīd al-Dīn features in biographical dictionaries as a latter-day Brother of Purity, indeed the veritable reincarnation of Sharaf al-Dīn Yazdī himself. For his scholarly oeuvre too was centrally

---

66 As testament to his great esteem, Ulugh Beg came to refer to Qūshchī as his adopted son.

67 Tehran, MS Majlis 72s (16th–17th c.), 1b–2b; Tehran, MS Majlis 16397 (undated), 17b–18a; ed. Sédillot 285–291.

68 On this new *zīj* as a definitive replacement for that of Ulugh Beg see Khan Ghori, Zīj Literature 34–36. Note, however, this study's dismissal of the remarkable lettrist section of the *Shāhjahānī Star Tables*' introduction, translated below, as merely treating of "numerical 'affinities' between the content of this *zīj* and the royal names and titles"—and hence of no scientific relevance whatsoever (35).

focused on historiography, lettrism, logogriphs and astronomy-astrology; and he was the star student of the preeminent Iranian polymath and occultist Mīr Fatḥ Allāh Shīrāzī (d. 998/1589), who was credited with the introduction of Islamic (read: Neoplatonic) philosophy to Akbar's India.[69] As a prominent Indian occult scientist of the day, Farīd al-Dīn was thus invited to produce one of the several natal and accession horoscopes designed as scientific proof for Shāhjahān's ontological equivalence to Temür and superiority to his great-grandfather Akbar.[70] On the strength of this commission, he was further asked to prepare a new *zīj* that would adapt the *Zīj-i jadīd-i sulṭānī* to Akbar's new Divine (*ilāhī*) calendar; and it was so impressive that a Sanskrit translation was commissioned shortly thereafter from the Brahmin scholar Nityānanda, who called it, significantly, the *Romakasiddhānta* ('Roman Star Tables').[71]

Nevertheless, Shāhjahān's primary focus as royal patron was not astronomy-astrology, unlike his predecessors Akbar and Humāyūn, but rather the arts of the book, especially painting, as well as architecture. He assiduously and rigorously oversaw his imperial painters in the production of ideologically talis-

69 Thus his biography in ʿAbd al-Ḥayy Ḥasanī's (d. 1923) *Nuzhat al-khawāṭir wa-bahjat al-masāmiʿ wa-l-nawāẓir* ('Minds' Diversion, Joys to See and Hear') v, 315–316, no. 504:

> The most learned shaykh and eminent litterateur Farīd al-Dīn b. Ibrāhīm al-Dihlawī al-Munajjim was without peer in his day in the mathematical disciplines. He was born and grew up in Delhi, and he first learned from his father, then Shaykh Niẓām al-Dīn al-Nārnūlī (God have mercy on him), with whom he studied various scholastic books; he then joined himself to Shaykh Fatḥ Allāh Shīrāzī and learned from him until he surpassed his peers in geometry, astronomy, astrology, numerology (*al-aʿdād*), the comprehensive prognosticon (*al-jafr al-jāmiʿ*) and many of the occult disciplines (*al-funūn al-gharība*), being particularly adept with star tables and talismans. As a luminary of these sciences he thus stood singular, as attested to by all scholars, and famed throughout the lands of India.
>
> Amir ʿAbd al-Raḥīm b. Bayram Khan made him part of his retinue and *ṣadr* of his army in the year 1006 [1598]. He was of praiseworthy comportment and extremely pious, spending his days in rigorous observation of worship and charity, of great virtue and abstemiousness, as reported in the Raḥīmian ['Record of] Noble Deeds' (*Maʾāṣir-i Raḥīmī*) [of ʿAbd al-Bāqī Nihāwandī]. He authored astonishing works and amazing (*gharīb*) inventions in such fields as riddles (*lughz*), history and lettrism (*jafr*), and most notably the *Shāhjahānī Star Tables*, which he produced in the year 1038 [1629] during the reign of Shāhjahān son of Jahāngīr. Therein he expended the utmost effort in editing the tables [of Ulugh Beg] and making them easier to use, fixing the defects in older usages and establishing rigorous and reliable principles on which to extrapolate the issues [of astronomy-astrology]. Its incipit: "Limitless praise is due God, the Geometer of Whose encompassing power" etc. He died in the year 1039 [1630], as recorded in the *Ṭabaqāt-i Shāhjahānī* ('Shāhjahānī Generations') [of Muḥammad Ṣādiq].

70 Pingree, The *Sarvasiddhāntarāja* 269.

71 Ibid. 269–270.

manic art,[72] and constructed a record 35 palaces (sg. *dawlatkhāna*) and garden residences (sg. *bāgh*)—"the largest extant body of palaces built by a single patron" in the whole of the Islamic architectural record.[73] The Tāj Maḥal, apotheosis of Timurid imperial architecture, thus epitomizes Shāhjahān's larger project to aggressively refashion—contemporaries might poetically have said *rewrite*—the Indian visual imagination and built landscape alike. Likewise, that it has as an important influence the tomb of Ulugh Beg himself is salient here.[74]

In the face of modern scientism, however, it must be emphasized that Shāhjahān's aesthetic patronage program must be considered scientific as well to the extent it was in fact *talismanic*—talismans being a standard technological application of natural-mathematical science from the 10th century onward. But here there be historiographical dragons.

Shāhjahān, it is true, did not go so far as to erect a Talismanic House (*khāna-yi ṭilism*) after the manner of Humāyūn—an astrologer himself, and (in)famous for his devotion to the occult sciences—, nor did he conduct court ritual in emphatically astral-magical terms after the manner of Akbar, the most successful millennial sovereign of Afro-Eurasia in the 16th century.[75] But to compete with his glorious grandfather, Shāhjahān would need to enter the polo pitch that is the field of astronomy-astrology with considerable force. The lettrist *Zīj-i Shāhjahānī* may therefore be considered as a royal throwing down of the ideological-*cum*-scientific gauntlet, and therefore expressive in some measure of royal authorial intent—hence its pregnant alternative title, *Kārnāma-yi ṣāḥib-qirān-i s̱ānī* ('Workbook of the Second Lord of Conjunction').

A translation of the key third section of the introduction follows, wherein Dihlavī presents a *tour de force* of astrological-lettrist historiography after the model set by his fellow New Brother Yazdī; most notably, it argues for Shāhjahān's ontological equivalence to Temür, on the one hand, and to the solar, lunar and Hindu calendars simultaneously, on the other (Yazdī rather argued for Temür as historical manifestation of the Quranic *ALM*):[76]

> Section three: On the numerological correspondences employed and identified in this auspicious New Zij (*zīj-i jadīd-i saʿīd*) dedicated to Shāh-

72 Moin, *The Millennial Sovereign* ch. 6.

73 Koch, The Taj Mahal 128.

74 Hoag, The Tomb of Ulugh Beg.

75 Koch, *The Planetary King*. On Mughal astral-magical court ritual generally see Orthmann, Court Culture; Moin, *The Millennial Sovereign*; Truschke, Solar Cosmology.

76 Melvin-Koushki, In Defense 388–391.

jahān [= 365], Ruler of India (*kāmrān-i Hind*) [= 371]—nay, Ruler of the World (*kāmrān-i jahān*) [= 371]—and Lord of Conjunction (*ṣāḥib-qirān*) [= Shah Khurram].[77]

It is evident that the first title (?) corresponds numerologically with the days of the lunar year (354), and that the second title (Shāhjahān) corresponds to the days of the solar year (365); the third and fourth titles [Ruler of India (371), Lord of Conjunction (452)] correspond to the *tata*s (?) of the Indian year. The *bayyināt*[78] of *ṣāḥib-qirān*, as laid out in the following table—

zubur	*Ṣ*	*A*	*Ḥ*	*B*	*Q*	*R*	*A*	*N*
bayyināt	*AD*	*LF*	*A*	*AF*	*A*	*A*	*LF*	*WN*

—likewise equal 365, which corresponds to the value of the blessed name Shāhjahān.

> In the world a *Lord of Conjunction* becomes the *Lord of the World*
> for the *bayyināt* of the first make manifest the second.

A further remarkable instance of meaningful synchronicity is the fact that the *bayyināt* of the glorious name of the holy First Lord of Conjunction (*ṣāḥib-qirān-i avval*), as laid out in this table—

T	*Y*	*M*	*W*	*R*
A	*A*	*YM*	*AW*	*A*

—equal 60, whose letter is three, the sign of the King of Heaven (*shāh-i falak*, i.e., the Sun) as represented in almanacs (*taqvīm*). Its revolution takes 365 days, the value of the name of the holy Second Lord of Conjunction (*ṣāḥib-qirān-i s̱ānī*); and what is even more remarkable is the fact that the *bayyināt* of the most noble and holy [name] Khurram Padishah also

77 I.e., Shāhjahān's pre-accession princely name.

78 When performing *taksīr* (cognate to Hebrew *temurah*), which involves the separation of the letters of a name or word and the writing out of the letternames in full, then the elimination of repeated letters (e.g., Aḥmad → *AḤMD* → *ALFḤAMYMDAL* → *ALFḤMYDL*), the term *zubur* refers to the first letters in the full letternames (e.g., the *A* in *ALF*) and *bayyināt* to the remaining letters (e.g., *LF* in *ALF*), with Q 3:184 and 35:25 as prooftext.

equal 365. At the moment of that holy and felicitated eminence's birth, moreover, the Ascendant was in Libra. Furthermore, the *bayyināt* of the most noble and holy name correspond to those of [the term] *zīj*, as is laid out in the following two tables:

zubur	*Z*	*Y*	*J*	*zubur*	*Kh*	*R*	*M*
bayyināt	*A*	*A*	*YM*	*bayyināt*	*A*	*A*	*YM*

This most noble and holy name has therefore been deployed in this auspicious New Zij as the Balance (*mīzān*, i.e., Libra) for all its calculations, including the mean motions (*awsāṭ*) of all the stars, their equations (*taʿdīlāt*), their longitudes (*taqāvīm*), etc., as will be discussed in the relevant sections below—indeed, even the calculation of years in chronicles is dependent thereon.

According to the dictum *Names are heaven-sent*, then, this most noble and holy name, generated by that holy eminence's auspicious horoscope, casts down from heaven its rays upon the realm of manifestation, like unto the Sun itself. It does so in this wise: The tenth house, i.e., the house of power and rule, is in Cancer (*saraṭān*), the sign of the Moon; Jupiter is here. Thus the *R* of *saraṭān* (*SRṬAN*), the sign of the Moon, may be replaced with *L*, the sign of Jupiter (*ZḤL*), to produce *sulṭān* (*SLṬAN*). The seventh house, moreover, is in Aries, and Mars (*MRYKh*) is there in its own sign; and the letters of its name, reordered according to the *BTTh* sequence, produces a nominal analogy with the most noble and holy name Khurram (*KhRM*), Joyous, with the *Y* as remainder of the word *MRYKh*, the sign of the tenth, being placed after the word *SLṬAN*, itself produced by the tenth house and associated therewith. When added to the phrase *Sulṭān Khurram* (= 990), the whole equals 1,000—the year of that holy eminence's felicitous birth. Furthermore, the fourth house is in Capricorn, and therein the Sun and Mercury are in exaltation, all of which indicates long life and stable power and rule. This is among the most subtle of all subtleties (*az alṭaf-i laṭāyif*) and the most remarkable of all meaningful synchronicities (*az ʿajībtarīn-i ittifāqāt-i ḥasana*), truly an embodiment of the dictum *What no eye has seen, nor ear heard, nor has entered into the heart of man …*

Also, the numerological value of the first letter of both his blessed name (Shāhjahān) and his blessed honorific (Shihāb al-Dīn), namely *Sh*, is 30. And the letters of his blessed name listed without repetition—

ShAHJN—together give 359, the antecedent (*muqavvim*) of [360], a full cycle (*dawr*). (The antecedent of each number is that which is one less than that number, as four to five or five to six.) In the Speech of the All-Knowing Sovereign, the phrase *Exalter of degrees, Possessor of the Throne* (Q 40:15) alludes to the degrees of the celestial sphere. The form of *Sh* (*ShYN*), moreover, has the *bayyināt* of [*YN*, equivalent to] *S* (*SYN*), whose lettername value is 120—the number of diametrical sections (*ajzā-yi quṭriyya*)—and whose *zubur* is 60—the number of arc minutes, seconds, thirds, fourths, etc. (That is, each arc degree has 60 arc minutes, and each arc minute has 60 arc seconds, and each second has 60 thirds, and each third has 60 fourths, and so on.) The value of the word *jahān* (*JHAN*), incorporated in his blessed name, is 59, with the same applying to *Hind* (*HND*); and 59 is the antecedent of 60. When the units letters of the phrase *vālī-yi vāliyān Shāhjahān khallada mulkahu va sulṭānah* (*WALYWALYAN-ShAHJHANKhLDMLKHWSLṬANH*) are extracted (*WAWAAAHJHADHW-ṬAH*), they too equal 60. The units value of Shāhjahān (*ShAHJHAN* > *AHJHA*) is 15, while that of *khallada mulkahu va sulṭānah* (*DHWṬAH*) is 30, corresponding to what is laid out on the page on stellar equations (*taʿdīlāt-i kavākib*) as to equation at 30 degrees of longitude and 15 degrees of latitude (as the word longitude (*ṭūl* = *ṬWL* = *ṬW* or 15 + *L* or 30) itself suggests). The units value of the undotted letters in the blessed name (*AHHA*) also corresponds to the number of zodiac signs, i.e., 12, constituting a complete cycle. Furthermore, the celestial degrees are divided into four quadrants (*arbāʿ*) [of 90 arc degrees each], this so as to construct a corrected arc (*qaws-i munaqqaḥ*) for determining the sine (*jayb*) and other necessary operations, which corresponds to the following four terms employed in encomium of that holy eminence: king (*malik* = *MLK*) [90], felicitated (*masʿūd* = *MSʿWD*) [180], munificent (*karīm* = *KRYM*) [270], exalted (*rafīʿ* = *RFYʿ*) [360]. There are in addition to these a number of other correspondences that have been noted in this book's proem *maṣnavī*.

It is likewise evident that in this auspicious New Zij the phrase "the most noble and holy name" (*ism-i ashraf-i aqdas* = *ASMAShRFAQDS*) refers to the name Khurram (*KhRM*), for the tens and hundreds value of the letters of the first (*SMShRFQS*) is equivalent to the value of the second (840); and the phrase "the blessed name" (*ism-i mubārak* = *ASMMBARK* = 364) refers to the name Shāhjahān (*ShAHJHAN* = 365), for the first is the antecedent of the second. And given that the numbers employed in this auspicious zij exactly correspond to the noble names of that holy eminence in whose sacred and exalted name it has been composed, as has been discussed above, and that the letters listed in this zij refer to those numbers, this preface has been written [by way of explanation]—despite the fact that [such correspondences] may not seem to have much bearing on [the science of] the stars (*nujūm*). For the points made here are subtle and occult (*nikāt-i laṭīfa u gharība*), and can be grasped by none but those who possess the intelligence to perceive the felicitating grandeur of the holy Shadow of God and be blessed thereby, magnifying it and counting it a wonder and marvel of the age (*az ʿajāyib u gharāyib-i rūzgār*). But God knows best.[79]

4 Dārā Shukūh, Sanskritist-Saint, Scientifically Occidentalizes the Orient

More historiographical ink has been spilled on the last of our Timurid royal authors than the other three combined; even the celebrated sultan-astronomer Ulugh Beg cannot compare.[80] For unlike his dynastic predecessors, Dārā Shukūh's self-identification as an otherworldly Sufi allows us to slot him into one side of our neat Religion vs. Science binary. Even more problematically, his later romantic image as a radically ecumenical but ill-starred heir to the Mughal throne, in stark and eternal opposition to his triumphant brother, Awrangzēb, epitome of militant Islamic chauvinism, makes him strategically indispensable to the nationalist Islam-Hinduism binary that now politically structures the Subcontinent.[81] But if we set aside these anachronistic, Neomanichean narratives, and restore Dārā Shukūh to his mid-17th-century Persianate context, a rather different picture emerges of this failed philosopher-king—one somewhat more *occult-scientific*.

This feature of his imperial-intellectual project, it must be emphasized, was guaranteed by the Sufi-prince's commitment to Ibn ʿArabī (d. 638/1240) in particular—the fountainhead of lettrist theory for the New Brethen of Purity and their many heirs. (He exhibits little interest in lettrist praxis, however, particularly the talismanic harnessing of the divine names, epitomized rather by the Sufi-mage Aḥmad al-Būnī (d. between 622–630/1225–1233).) Most importantly, as a universal occult science transcending the binary of the sciences rational and religious, Hellenic and Islamic, the brand of Ibn Turkian lettrism

79 British Library, MS Or. 372 (17th c.), ff. 4^b–5^a.

80 This disproportionality is suggested by a survey of *Index Islamicus*.

81 Gandhi, *The Emperor Who Never Was*, intro.; Kinra, Infantilizing Bābā Dārā; Truschke, *Aurangzeb*.

embedded, as we have seen, in Timurid imperial ideology from its inception emphasized the philosophical but also sociopolitical need to *marry all opposites*.[82] The lettrist doctrine of the *coincidentia oppositorum* (*majmaʿ al-aḍdād*) was equally salient for the Neopythagoreans of contemporary Latin Christendom, to be sure (Nicholas Cusanus (d. 1464), Ibn Turka's later contemporary, coined the Latin term); but the bloody, irreparable rupture that was the Protestant Reformation ensured that this philosophical ideal would never be sociopolitically realized in western Europe.[83] In extreme contrast, this ideal was consciously, even aggressively, pursued, and at least partially and occasionally realized, by Temür's heirs in India.[84]

Like Akbar before him, Dārā Shukūh's overriding concordist project was thus the marriage of the Islamic and the Indic opposites, so constructed for this purpose—and as an author, translator, artist and patron he so successfully married them that even inner-Hindu discourse was shaped by his Ibn ʿArabian monistic vision over the following centuries.[85] This project is epitomized by the very title of one of his most influential works, the *Majmaʿ al-baḥrayn* ('Meeting of the Two Seas'), which "offers a distillation of Indic and Islamic concepts [mapping] the topography of the universe, the self, and the afterlife, encompassing the macrocosm as well as the microcosm";[86] it includes, not incidentally, a dual investigation into the divine names in Arabic and Sanskrit.[87]

But the saint-prince's project was fully actualized in the *Sirr-i Akbar* ('Supreme Secret'), his Persian translation and anthologization of the Upaniṣads, which exerted considerable textual influence on subsequent Islamic and Indic discourses alike. It is emblematic of the same taḥqīqist ethos that drove the (occult-)scientific projects of Iskandar Sulṭān and Ulugh Beg before him, here applied not to astronomy-astrology but to comparative religion.[88] And where his Timurid Neopythagorean forebears succeeded in mathematizing the science of the stars, Dārā succeeded in *occidentalizing the orient*—this by con-

82 Melvin-Koushki, Imperial Talismanic Love.

83 On Cusanus see Albertson, *Mathematical Theologies*; on European Renaissance parallels more generally, see Joost-Gaugier, *Pythagoras*.

84 On contradiction and ambiguity as guiding themes in Islamicate societies, see Ahmed, *What Is Islam?*; Bauer, *Die Kultur der Ambiguität*.

85 Gandhi, Mughal Self-Fashioning 283–291. For a critique and rejection of the notion of a stable "Hinduism" during this period, however, with reference to Dārā Shukūh's oeuvre, see ibid. and Gandhi, *The Emperor Who Never Was*; and Ernst, Muslim Studies esp. 185–187.

86 Gandhi, Mughal Self-Fashioning 137.

87 This section is translated in ibid. 313–314.

88 On the term "taḥqīqism" as tantamount to early modernity, see Melvin-Koushki, *Taḥqīq vs. Taqlīd*.

stellating and then casting Hinduism as a primordial form of monotheism, and the Quran a modern epitome of ancient and encompassing Vedic scripture. In contrast to contemporary Latin Christian recastings of indispensable Muslim luminaries as Europeans, born of chauvinism, Dārā's project is concerned to establish the total parity of East and West—a move that may be styled Oriento-Occidentalism.[89]

That our royal author chose to rhetorically frame the *Supreme Secret* in explicitly *lettrist* terms is therefore of the highest significance for understanding his strategies of specifically Timurid self-fashioning. This Dārā does in the first lines of the preface, translated below; as with Ulugh Beg's own preface to his equally epochal zij, such Persian rhetorical flourishes are highly coded and repay close analysis. Here two major lettrist themes are invoked: the *B* of the *basmala* as matrix of both the Quran and the cosmos, and the Name of God, which themes together constitute *scientific* proof for the universality of Quranic-Vedic authority—and hence Islamic-Indic Empire.

At the same time, neither here nor elsewhere does Dārā discuss the practical, which is to say magical, implications of such high lettrist theory, and declines to pair the science with astrology after the manner of his Timurid predecessors. He has little time, that is, for occult science, much less occult technology—an apparent disinclination that doubtlessly reflects his fatal lack of political pragmatism more generally. But that disinclination was not total, as he hints at the end of the same preface: the Sufi-prince declares that he only undertook this unprecedented project at the instance of a bibliomantic—i.e., lettrist—operation, which returned, significantly, a passage headed and hence guaranteed by *muqaṭṭaʿāt*, those elemental fragments of the primordial cosmic scripture that was first temporally expressed in the *hidden book* (Q 56:78: *kitāb maknūn*) that is the Upaniṣads. That even such a putatively otherworldly individual as Dārā Shukūh thought it only natural to engage in such operations is further evidence (were any still needed) for the ubiquity of both lettrist theory and lettrist practice among the scholarly and courtly elites of the early modern Persian cosmopolis.

The two relevant passages follow:

> True praise is essential—it strikes dumb the tongue; among its ancient mysteries is the *B* of *bismi Llāh* in all heavenly Books. The Quranic [sura] *The Praise* (Q 1)—the Mother of the Book—refers to His Supreme Name,

89 See ibid. 198, 201; Melvin-Koushki, India Is Magic.

> which Name incorporates all angels and heavenly Books and prophets and saints ....[90]
>
> Having ascertained that the *hidden book* refers to this ancient Book [scil., the Upaniṣads], all unknowns and misunderstandings became known to and rightly understood by this wretch. And before beginning to translate, I performed a Quranic divination, and the result was the sura *The Elevations* (*al-Aʿrāf*), which opens: *ALMṢ. A Book sent down to you—let not your breast be constricted because of it—to warn thereby, and as a reminder to believers* (Q 7:1–2).[91]

Read in tandem with the introduction to the *Zīj-i Shāhjahānī*, in sum, the preface to the *Sirr-i Akbar* stands brief but eloquent testament to the continued centrality of letter science and star science to Timurid universalist imperialist ideology, two centuries after its first formulation by the ambitious philosopher-kings Iskandar Sulṭān and Ulugh Beg and their faithful scholar-scientist ideologues Ibn Turka and Yazdī.

Conclusion: Imperially Reading and Writing the World

Such textual continuity suggests that the Neopythagorean cosmology of our Timurid-Mughal royal actors and authors must be taken far more seriously than it has been to date. That they held the cosmos itself to be a mathematical text to be riddled by scientists of stars and letters alone explains the contours of their remarkable intellectual-imperial projects, which serve in turn as chiefest index of the advent of a specifically Western—because Pythagorean-Platonic—scientific early modernity. To these ambitious exponents of the Two Books doctrine, the complex of Persian literary tropes styling space and time as Quranic manuscript pages to be calligraphed and illuminated could be, and was sociopolitically, taken quite literally—which is to say, *lettrarily*.[92]

For if the world is indeed a mathematical-linguistic text demanding to be read, it also demands to be *written*, preferably in marble. May we thus not style Shāhjahān himself author of the unparalleled cosmographical logogriph (*muʿammā*)—which poetical genre was made a mainstay of lettrist practice precisely by the Timurid Brethren of Purity—that is the Tāj Maḥal?[93]

---

90 Thus Tehran, MS Millī 816408 (undated), p. 1; the edition by Chand and Nāʾīnī drops the first clause (ed. iii).

91 Ibid. 5–6.

92 I propose this term in Melvin-Koushki, The New Brethren.

93 Yazdī, naturally, was the first in the Arabo-Persian literary tradition—and indeed the Western as a whole—to explicitly theorize the humble *muʿammā* as a crucial skill in the

Bibliography

Primary Sources

Dārā Shukūh, *Sirr-i Akbar*, ed. Tārā Chand and Muḥammad-Riżā Jalālī Nāʾīnī, Tehran 1957.

Tehran, MS Millī 816408 (undated).

Dihlavī, Farīd al-Dīn Masʿūd b. Ibrāhīm, *Kārnāma-yi ṣāḥib-qirān-i s̱ānī, yā Zīj-i Shāhjahānī*. British Library, MS Or. 372 (17th c.).

Ḥasanī, ʿAbd al-Ḥayy b. Fakhr al-Dīn, *Nuzhat al-khawāṭir wa-bahjat al-masāmiʿ wa-l-nawāẓir*, 8 vols. in 4, Hyderabad 1947–1970.

Iskandar Mīrzā, *Dībācha-yi Jāmiʿ al-sulṭānī*, in Sharaf al-Dīn ʿAlī Yazdī, *Munshaʾāt*, ed. Īraj Afshār, Tehran 1387 Sh./2008, 207–211.

Jurjānī, Sayyid Sharīf, *al-Taʿrīfāt*, ed. ʿAbd al-Raḥmān ʿUmayra, Beirut 1407/1987.

Majlisī, Muḥammad Bāqir, *Biḥār al-anwār*, 110 vols., Beirut 1403/1983.

Ulugh Beg, Preface, *Zīj-i jadīd-i sulṭānī*. Tehran, MS Majlis 72s (16th–17th c.), 1b–2b; MS Majlis 16397 (undated), 17b–18a, in L.-P.-E.A. Sédillot (ed.), *Prolégomènes des tables astronomiques d'Oloug-Beg: publiées avec notes et variantes, et précédés d'une introduction*, Paris 1847, 285–291. Trans. in idem, *Prolégomènes des tables astronomiques d'Oloug-Beg: traduction et commentaire*, Paris 1853, 1–6.

Secondary Literature

Ahmed, Shahab, *What Is Islam? The Importance of Being Islamic*, Princeton 2017.

Albertson, David, *Mathematical Theologies: Nicholas of Cusa and the Legacy of Thierry of Chartres*, New York 2014.

---

lettrist's technical repertoire: an immediate, poetic-mathematical means of analyzing a person's name in order to discern their character, perhaps even their fate (Melvin-Koushki, The Quest 379–389; Binbaş, *Intellectual Networks* 48, 81–89). On this reading—or rather riddling—of the Tāj Maḥal, see Parodi, The Distilled Essence. Likewise suggestive is the fact that the two sides of Mumtāz Maḥall's (d. 1040/1631) lower cenotaph feature the 99 Names of God in their entirety—the primary focus, of course, of lettrist magic by the early modern period (Begley and Desai, *Taj Mahal* 240–241). More broadly, as Ebba Koch notes, the Tāj Maḥal "is 'built architectural theory,' which can be read almost like a literary text once we have mastered the grammar and vocabulary of the architectural language … We note here the purest expression of a consistent formal systematization characteristic of the entire art of Shah Jahan; it represents a distinctive and outstanding contribution specific to this period" (The Taj Mahal 137–138). Most significantly, it was designed to embody the *coincidentia oppositorum* through the construction, unfortunately unrealized, of its equal and opposite in black marble across the river Yamuna. On the similarly revolutionary nature of Mughal garden culture, with comparisons to Renaissance Europe, see Koch, Carved Pools; cf. Subtelny, *Le monde est un jardin*. On the preeminent Safavid sage-mage Shaykh Bahāʾī's (d. 1030/1621) revolutionary redesign of Isfahan along similarly Pythagorean lines, see Melvin-Koushki, The New Brethren.

Aubin, Jean, Le mécénat timouride à Chiraz, in *Studia Islamica* 8 (1957), 71–88.

Azzolini, Monica, *The Duke and the Stars: Astrology and Politics in Renaissance Milan*, Cambridge, Mass. 2013.

Babayan, Kathryn, *Mystics, Monarchs, and Messiahs: Cultural Landscapes of Early Modern Iran*, Cambridge, Mass. 2002.

Bauer, Thomas, *Die Kultur der Ambiguität: eine andere Geschichte des Islams*, Berlin 2011.

Begley, W.E. and Z.A. Desai, *Taj Mahal: The Illumined Tomb: An Anthology of Seventeenth-Century Mughal and European Documentary Sources*, Cambridge, Mass. 1989.

Bdaiwi, Ahab, Late Antique Intellectualism in Medieval Islam: The Shiraz Circle and the Revival of Ancient and Islamic Knowledge, *Journal of Late Antique, Islamic and Byzantine Studies* 2, nos. 1–2 (2023), 128–171.

Binbaş, İ. Evrim, *Intellectual Networks in Timurid Iran: Sharaf al-Dīn ʿAlī Yazdī and the Islamicate Republic of Letters*, Cambridge 2016.

Binbaş, İ. Evrim, Timurid Experimentation with Eschatological Absolutism: Mīrzā Iskandar, Shāh Niʿmatullāh Walī, and Sayyid Sharīf Jurjānī in 815/1412, in Orkhan Mir-Kasimov (ed.), *Unity in Diversity: Patterns of Religious Authority in Islam*, Leiden 2014, 277–303.

Binbaş, İ. Evrim, The Anatomy of a Regicide Attempt: Shāhrukh, the Ḥurūfīs, and the Timurid Intellectuals in 830/1426–1427, in *Journal of the Royal Asiatic Society* 23 (2013), 391–428.

Blake, Stephen P., *Astronomy and Astrology in the Islamic World*, Edinburgh 2016.

Brack, Jonathan Z., Mediating Sacred Kingship: Conversion and Sovereignty in Mongol Iran, Ph.D. diss., University of Michigan, 2016.

Brack, Jonathan Z., *An Afterlife for the Khan: Muslims, Buddhists, and Sacred Kingship in Mongol Iran and Eurasia*, Berkeley 2023.

Brentjes, Sonja, *Teaching and Learning the Sciences in Islamicate Societies (800–1700)*, Turnhout 2018.

Brentjes, Sonja, The Mathematical Science in Safavid Iran: Questions and Perspectives, in Denis Hermann and Fabrizio Speziale (eds.), *Muslim Cultures in the Indo-Iranian World during the Early-Modern and Modern Periods*, Berlin 2010, 325–402.

Bürgel, Johann Christoph, Occult Sciences in the *Iskandarnameh* of Nizami, in Kamran Talattof and Jerome W. Clinton (eds.), *The Poetry of Nizami Ganjavi: Knowledge, Love, and Rhetoric*, New York 2000, 129–139.

Caiozzo, Anna, The Horoscope of Iskandar Sultān as a Cosmological Vision in the Islamic World, in Günther Oestmann, H. Darrel Rutkin and Kocku von Stuckrad (eds.), *Horoscopes and Public Spheres: Essays on the History of Astrology*, Berlin 2005, 115–144.

Chann, Naindeep, Lord of the Auspicious Conjunction: Origins of the *Ṣāḥib-Qirān*, in *Iran and the Caucasus* 13 (2009), 93–110.

Elverskog, Johan, The Mongols, Astrology and Eurasian History, in *The Medieval History Journal* 19/1 (2016), 130–135.

*EI*³ = *Encyclopaedia of Islam, Three*, Leiden 2007–.

Ernst, Carl W., Muslim Studies of Hinduism? A Reconsideration of Arabic and Persian Translations from Indian Languages, in *Iranian Studies* 36/2 (2003), 173–195.

Fazlıoğlu, İhsan, The Samarqand Mathematical-Astronomical School, in *Journal for the History of Arabic Science* 14 (2008), 3–68.

Fazlıoğlu, İhsan, İlk Dönem Osmanlı İlim ve Kültür Hayatinda İhvanu's-Safâ ve Abdurrahman Bistâmî, *Dîvân: İlmî Araştırmalar* 1 (1996), 229–240.

Flatt, Emma, The Authorship and Significance of the *Nujūm al-ʿulūm*: A Sixteenth-Century Astrological Encyclopedia from Bijapur, in *Journal of the American Oriental Society* 131/2 (2011), 223–244.

Fleischer, Cornell H., Ancient Wisdom and New Sciences: Prophecies at the Ottoman Court in the Fifteenth and Early Sixteenth Centuries, in Massumeh Farhad and Serpil Bağcı (eds.), *Falnama: The Book of Omens*, London 2009, 232–243.

Gandhi, Supriya, *The Emperor Who Never Was: Dara Shukoh in Mughal India*, Cambridge, Mass., 2020.

Gandhi, Supriya, Mughal Self-Fashioning, Indic Self-Realization: Dārā Shikoh and Persian Textual Cultures in Early Modern South Asia, Ph.D. diss., Harvard University, 2011.

Håkansson, Håkan, *Seeing the Word: John Dee and Renaissance Occultism*, Lund 2001.

Hallum, Bink, New Light on Early Arabic *Awfāq* Literature, in Liana Saif, Francesca Leoni, Matthew Melvin-Koushki and Farouk Yahya (eds.), *Islamicate Occult Sciences in Theory and Practice*, Leiden 2021, 57–161.

Hayton, Darin, *The Crown and the Cosmos: Astrology and the Politics of Maximilian I*, Pittsburgh 2016.

Hoag, John D., The Tomb of Ulugh Beg and Abdu Razzaq at Ghazni, a Model for the Taj Mahal, in *Journal of the Society of Architectural Historians* 27/4 (1968), 234–248.

Huffman, Carl A., *Archytas of Tarentum: Pythagorean, Philosopher and Mathematician King*, Cambridge 2005.

Joost-Gaugier, Christiane, *Pythagoras and Renaissance Europe: Finding Heaven*, Cambridge 2014.

Karjoo-Ravary, Ali, Becoming a 'King of Islam': The Imperial Project of Qadi Burhan al-Din of Sivas (1345–1398 CE), Ph.D. diss., University of Pennsylvania, 2018.

Kennedy, Edward S., The Heritage of Ulugh Beg, in idem, *Astronomy and Astrology in the Medieval Islamic World*, Aldershot 1998, ch. XI.

Khan Ghori, S.A., Development of Zīj Literature in India, in S.N. Sen and K.S. Shukla (eds.), *History of Astronomy in India*, New Delhi 1985, 21–48.

Killeen, Kevin and Peter J. Forshaw (eds.), *The Word and the World: Biblical Exegesis and Early Modern Science*, Houndsmill, Basingstoke 2007.

Kinra, Rajeev, Infantilizing Bābā Dārā: The Cultural Memory of Dārā Shekuh and the Mughal Public Sphere, in *Journal of Persianate Studies* 2 (2009), 165–193.

Koch, Ebba, *The Planetary King: Humayun Padshah, Inventor and Visionary on the Mughal Throne*, Ahmedabad 2022.

Koch, Ebba, Carved Pools, Rock-Cut Elephants, Inscriptions, and Tree Columns: Mughal Landscape Art as Imperial Expression and Its Analogies to the Renaissance Garden, in Mohammad Gharipour (ed.), *Gardens of Renaissance Europe and the Islamic Empires: Encounters and Influences*, University Park 2017, 185–212.

Koch, Ebba, The Taj Mahal: Architecture, Symbolism, and Urban Significance, in *Muqarnas* 22 (2005), 128–149.

Lemay, Richard, *Abu Maʿshar and Latin Aristotelianism in the Twelfth Century: The Recovery of Aristotle's Natural Philosophy through Arabic Astrology*, Beirut 1962.

Markiewicz, Christopher A., The Crisis of Rule in Late Medieval Islam: A Study of Idrīs Bidlīsī (861–926/1457–1520) and Kingship at the Turn of the Sixteenth Century, Ph.D. diss., University of Chicago, 2015.

Melvin-Koushki, Matthew, *India Is Magic: Orientalism and Oriento-Occidentalism in the Persianate Occult Enlightenment*, forthcoming.

Melvin-Koushki, Matthew, *The Occult Science of Empire in Aqquyunlu-Safavid Iran: Two Shirazi Lettrists and Their Manuals of Magic*, Leiden forthcoming.

Melvin-Koushki, Matthew, Toward a Neopythagorean Historiography: Kemālpaşazāde's (d. 1534) Lettrist Call for the Conquest of Cairo and the Development of Ottoman Occult-Scientific Imperialism, in Liana Saif, Francesca Leoni, Matthew Melvin-Koushki and Farouk Yahya (eds.), *Islamicate Occult Sciences in Theory and Practice*, Leiden 2021, 380–419.

Melvin-Koushki, Matthew, The New Brethren of Purity: Ibn Turka and the Renaissance of Neopythagoreanism in the Early Modern Persian Cosmopolis, in *Intellectual History of the Islamicate World* 11/2 (forthcoming 2023).

Melvin-Koushki, Matthew, Imperial Talismanic Love: Ibn Turka's *Debate of Feast and Fight* (1426) as Philosophical Romance and Lettrist Mirror for Timurid Princes, in *Der Islam* 96/1 (2019), 42–86.

Melvin-Koushki, Matthew, *Taḥqīq* vs. *Taqlīd* in the Renaissances of Western Early Modernity, in *Philological Encounters* 3/1 (2018), 193–249.

Melvin-Koushki, Matthew, Early Modern Islamicate Empire: New Forms of Religiopolitical Legitimacy, in Armando Salvatori, Roberto Tottoli and Babak Rahimi (eds.), *The Wiley-Blackwell History of Islam*, Hoboken 2018, 353–375.

Melvin-Koushki, Matthew, Persianate Geomancy from Ṭūsī to the Millennium: A Preliminary Survey, in Nader El-Bizri and Eva Orthmann (eds.), *Occult Sciences in Premodern Islamic Cultures*, Beirut 2018, 151–199.

Melvin-Koushki, Matthew, In Defense of Geomancy: Šaraf al-Dīn Yazdī Rebuts Ibn Ḫaldūn's Critique of the Occult Sciences, in Matthew Melvin-Koushki and Noah

Gardiner (eds.), *Islamicate Occultism: New Perspectives*, Special double issue of *Arabica* 64/3–4 (2017), 346–403.

Melvin-Koushki, Matthew, Powers of One: The Mathematicalization of the Occult Sciences in the High Persianate Tradition, in *Intellectual History of the Islamicate World* 5/1 (2017), 127–199.

Melvin-Koushki, Matthew, The Quest for a Universal Science: The Occult Philosophy of Ṣāʾin al-Dīn Turka Iṣfahānī (1369–1432) and Intellectual Millenarianism in Early Timurid Iran, Ph.D. diss., Yale University, 2012.

Melvin-Koushki, Matthew, The Delicate Art of Aggression: Uzun Hasan's *Fathnama* to Qaytbay of 1469, in *Iranian Studies* 44/2 (2011), 193–214.

Moin, A. Azfar, *The Millennial Sovereign: Sacred Kingship and Sainthood in Islam*, New York 2012.

Orthmann, Eva, Court Culture and Cosmology in the Mughal Empire: Humāyūn and the Foundations of the *dīn-i ilāhī*, in Albrecht Fuess and Jan-Peter Hartung (eds.), *Court Cultures in the Muslim World: Seventeenth to Nineteenth Centuries*, London 2014, 202–220.

Orthmann, Eva, Circular Motions: Private Pleasure and Public Prognostication in the Nativities of the Mughal Emperor Akbar, in Günther Oestmann, H. Darrel Rutkin and Kocku von Stuckrad (eds.), *Horoscopes and Public Spheres: Essays on the History of Astrology*, Berlin 2005, 101–114.

Parodi, Laura E., 'The Distilled Essence of the Timurid Spirit': Some Observations on the Taj Mahal, in *East and West* 50 (2000), 535–542.

Pingree, David, The *Sarvasiddhāntarāja* of Nityānanda, in Jan P. Hogendijk and Abdelhamid I. Sabra (eds.), *The Enterprise of Science in Islam: New Perspectives*, Cambridge, Mass. 2003, 269–284.

Popp, Stephan, Mughal Horoscopes as Propaganda, in *Journal of Persianate Studies* 9 (2016), 45–59.

Ragep, F. Jamil, Copernicus and His Islamic Predecessors: Some Historical Remarks, in *History of Science* 45 (2007), 65–81.

Ryan, Michael A., *A Kingdom of Stargazers: Astrology and Authority in the Late Medieval Crown of Aragon*, Ithaca 2011.

Saliba, George, *Islamic Science and the Making of the European Renaissance*, Cambridge, Mass. 2007.

Savādī, Fāṭima, Naqdī bar istidlāl-i Ruzinfild [Rosenfeld] dar bāb-i intisāb-i yak risāla-yi riyāżī bi Ulugh Bayg (d. 853 q), in *Majalla-yi Tārīkh-i ʿIlm* 4 (1384 Sh./2005), 85–103.

Schmidl, Petra G., Magic and Medicine in a Thirteenth-Century Treatise on the Science of the Stars, in Ingrid Hehmeyer, Hanne Schönig and Anne Regourd (eds.), *Herbal Medicine in Yemen: Traditional Knowledge and Practice, and Their Value for Today's World*, Leiden 2012, 43–68.

Şen, Ahmet Tunç, Astrology in the Service of the Empire: Knowledge, Prognostication, and Politics at the Ottoman Court, 1450s–1550s, Ph.D. diss., University of Chicago, 2016.

Subtelny, Maria E., *Le monde est un jardin: aspects de l'histoire culturelle de l'Iran médiéval*, Paris 2002.

Truschke, Audrey, *Aurangzeb: The Life and Legacy of India's Most Controversial King*, Stanford 2017.

Truschke, Audrey, Translating the Solar Cosmology of Sacred Kingship, in *The Medieval History Journal* 19/1 (2016), 136–141.

van Dalen, Benno, Ulugh Beg: Muḥammad Ṭaraghāy ibn Shāhrukh ibn Tīmūr, in Thomas Hockey et al. (eds.), *The Biographical Encyclopedia of Astronomers*, New York 2007, 1157–1159.

Varisco, Daniel Martin, *Medieval Agriculture and Islamic Science: The Almanac of a Yemeni Sultan*, Seattle 1994.

Whitehead, Alfred North, *Science and the Modern World: Lowell Lectures, 1925*, New York 1925.

Yates, Frances A., *Astraea: The Imperial Theme in the Sixteenth Century*, London 1975.